Image-Based Tissue Characterization of an Ex Ovo Pressure Model for Chronic Wounds

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Introduction

Unresolved inflammation disrupts the normal wound healing process, leading to chronic wounds. Among them, pressure ulcers (PU) develop when sustained mechanical loads induce vascular occlusion, resulting in tissue ischemia, hypoxia, and persistent inflammation [1]. The chick embryo chorioallantoic membrane (CAM) provides a highly vascularized, non-sentient *ex ovo* model. It enables direct assessment of vascular remodeling together with the associated tissue response in a real-time system [2].

This work aimed to develop an Ex *Ovo* Pressure Model based on the CAM assay, in which controlled pressure induces localized vascular occlusion to reproduce the early pathological events of PU formation. The model was evaluated through image and histological analysis.

Methods

***Ex ovo* model.** Fertilized chicken eggs were cultured using the *ex ovo* CAM protocol described by Kohli *et al*. [2]. Eggs were incubated in rotation from embryonic development day (EDD) 0 to EDD3, then transferred to ex ovo culture. Pressure-inducing (PI) materials were implanted on EDD7, samples were collected on EDD10.

PI materials. PDMS discs (Ø = 6 mm) were implanted on EDD7. Three mechanical loading conditions were generated by varying the disc thickness (H1, H2, and H3). PDMS was selected because of its biological inertness, allowing the isolated evaluation of the mechanical effect.

Image and histological analysis. At EDD10, vascular images were acquired using a 4K digital microscope, after which samples were processed for histological analysis. The quantitative image analysis workflow combines Ilastik and MATLAB. Images were preprocessed by defining regions of interest (ROIs), segmented through machine learning algorithms, and analyzed in MATLAB to quantify non-fractal vascular parameters.

The overall workflow is summarized in Figure 1.

Results

Microscopic inspection revealed pressure-dependent vascular alterations. Increasing mechanical loading causes progressive disruption of blood vessels, whereas newly formed vessels tended to circumvent the implant rather than grow underneath it.

The proposed quantitative workflow captured vascular morphological changes through complementary non-fractal parameters. While vascular area percentage, branch point (BP) density, vessel length, and vessel diameter quantify the loss of vascularization and angiogenic capability, tortuosity descriptors such as distance metric (DM) and total curvature (TC) characterize changes in vessel direction.

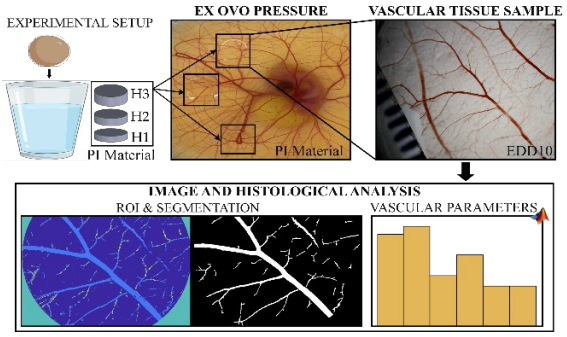


Figure 1: Overview of the workflow. Ex Ovo Pressure Model combines the ex-ovo CAM model with PI materials. At EDD10, samples were analyzed by imaging, to quantify the vascular parameters, and by histological analysis.

Table 1 illustrates the quantitative results generated by the image analysis workflow for a control case (C) and a load case (H1).

	Vascular area (%)	BP density (bp/mm^2)	Length density (mm/mm^2)	Median diameter (mm)	DM (-) TC (rad/mm)
C	7.40	1.81	2.12	0.037	1.25, 14.03
H1	4.85	0.065	0.62	0.084	1.97, 19.39

Table 1: Non-fractal vascular parameters quantified from a tissue control image (C) and a load image (H1).

Conclusions

The proposed *Ex Ovo* Pressure Model reproduces the early PU vascular alterations. Combining histology and quantitative vascular analyses provides a complete characterization of tissue response, supporting the use of this model to evaluate therapeutic strategies.

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References

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